

Khaled AlKharashi

IT System Administrator & Electronics

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Profile

Highly motivated IT professional with a solid background in **electronics, networking, VoIP systems, and IoT integrations**, supported by hands-on experience in technical support and infrastructure operations. Skilled in managing and troubleshooting **network devices, system hardware, and software environments**, as well as configuring and maintaining **LAN/WAN, routers, switches, and VoIP solutions**. Adept at diagnosing complex technical issues, performing preventive maintenance, and delivering reliable support for daily IT operations. With strong problem-solving skills and a foundation in **IoT and low-level technologies**, I bring a practical and analytical approach to system administration tasks. Known for working effectively under pressure, quickly resolving issues, and ensuring stable and secure IT environments.

Professional Experience

Network Support Engineer - VoIP implementer, Egypt Telecom, Xceed - Managed enterprise data and network infrastructure, diagnosing physical and logical faults and coordinating technicians for rapid issue resolution. - Configured, maintained routers, switches, terminals to ensure stable, high-performance connectivity. - Implemented and administered VoIP systems, including call center setups, VPN-based internal solutions, and SIP server configurations. - Integrated IP phones and gateways (Yeastar S/P-Series, etc.) with internal systems, external SIP trunks, and API providers. - Administered core network protocols: SIP Trunking, TCP/IP, DNS, DHCP, NAT, VLAN. - Performed Linux system administration. - VoIP platforms such as Asterisk, FreePBX, Avaya, 3CX, legacy PBX/UC systems.	10/2021 – 02/2023 Alexandria, Burj Al Arab
Electronics & DC Devices Maintenance Technician - Center Owner, Self-Employed - Repaired, diagnosed computer/laptop, motherboards, DC devices hardware/software. - Fixed power circuits, replaced ICs, SMD components, MOSFETs, regulators, capacitors. - Installed and optimized operating systems, drivers, troubleshooting software issues. - Performed PCB-level repair and fault tracing using multimeters and oscilloscopes. - Managed end-to-end operations, customer support, inventory, quality assurance. - Delivered more than 700+ successful repairs with high customer satisfaction.	05/2018 – 06/2020 Egypt
Electronics Maintenance Technician, Al Araby Group - Authorized Centers - Repaired and diagnosed DC electronic devices including receivers, audio/video equipment, power circuits, and PCB-based systems. - Performed component-level analysis, SMD, IC replacement, board-level troubleshoot. - Conducted end-to-end diagnostics, oscilloscopes, multimeters, signal testing tools. - Improved device stability by identifying repeated circuit failures and recommending component-level solutions.	01/2014 – 05/2018 Egypt

TECHNICAL TRAINING & PROJECTS

IoT, Embedded Systems Development Experience (Hands-on Real Word Projects) - ITI, Egypt, Mansoura, Information Technology Institution, Ministry of Communications and Information Technology – Internet of Things Track

- Designed, programmed embedded systems using Arduino, MikroC, PIC16F87XA microcontrollers, Proteus.
- Developed IoT prototypes integrating sensors, actuators, communication modules.
- Worked with digital/analog circuits, UART, I2C, and SPI communication protocols.
- Built small-scale IoT systems include measurement units, automation modules, microcontroller-based projects.
- Applied electronics repair experience to enhance embedded hardware design, debugging, and testing.
- Gained hands-on experience in edge-device programming, basic PCB design, and hardware integration.

IoT Smart Parking System, using proximity sensors to detect real-time parking availability and send status data to an IoT platform. The system updates a live parking map, showing occupied and free spots for faster navigation and efficient space management.

IoT Smart Lighting System, using a light sensor to automatically turn lights on or off based on ambient light levels. The system detects when brightness goes above or below a set threshold and adjusts lighting accordingly, improving energy efficiency for homes, offices, and public spaces.

IoT Smart Bag, system that tracks real-time location and open/close status using a light sensor. Status and location data are sent to an IoT platform, providing live monitoring and event history for enhanced security and traceability.

Motion Simulation, Real-time motion tracking system using acceleration sensors and IoT connectivity. The system transmits movement data to an IoT platform, providing live position updates and insights for applications like robotics, logistics, and asset monitoring.

IoT Tracking Locations, Real-time multi-location tracking and event-driven alerts powered by IoT sensor data. with monitor dashboard and locations history.

Desktop Console Application (Team Work), Console-based Library Management System in C with team collaboration. Supports Guests, Users, and Admins with authentication, CRUD operations via file I/O for persistent storage, and an interactive console GUI using cursor positioning and lightbar menus.

Skills

Electronics Engineering — Expert

- Hardware Troubleshooting & Board-Level Repairs
- DC Devices, Audio/Video Systems, Satellite Receivers
- SMD Soldering, Component Replacement
- Firmware Flashing, Device Testing, Circuit Diagnostics
- Electronics Datasheets, schematic diagram Interpretation

Software, API Integration (Web) — Expert

- Node.js, Python, C#, .NET, JAVA
- HTTP/HTTPS, MQTT, Microservices (Solid Basics)
- REST, Restful - SOAP APIs, Design Patterns
- MongoDB, MySQL, PostgreSQL, ORM/ODM
- Typescript, React & Next.js dashboards
- Authentication, Authorization (JWT), CRUD
- Git, Apache web server, JSON, XML

Embedded & IoT — Proficient

- Arduino, EmbeddedC - MikroC, PIC Microcontrollers (PIC16F87XA), Proteus, Low Level Programming
- Sensors & Actuators, IoT Prototyping
- Serial Communication: UART, I2C, SPI
- Circuit Design, Debugging, Power Supply Design
- Basic PCB Design & Microcontroller Programming

Network & VoIP — Competent

- Network Technical Support Engineer (ISP-level): Logical/physical network support, VOIP/Unified, Communications system implementation
- Protocols & Concepts: LAN/WAN, TCP/IP, DHCP, DNS, NAT, Routing & Switching, VLANs, VPNs, VoIP
- VoIP Support: SIP configuration, softphone setup, basic Asterisk/Traditional PBX, landlines, QoS troubleshooting
- Linux Administration, Virtualization, Shell, SSH Keys

Education

Technical Diploma (ITI Scholarship) , Information Technology Institution	07/2020 – 08/2021
Internet of Things & Embedded, Software Engineering, MERN, Cloud Platforms.	Egypt, Mansoura

Higher Diploma , Educational Technology	05/2015 – 06/2016
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Bachelor Degree , University of ART, Anthropology, Sociology	2010 – 2014
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Achievements

- Successfully repaired and restored hundreds of consumer electronics devices with accuracy and reduced repair cycle times.
- Developed multiple IoT mini-projects integrating microcontrollers and sensors during scholarship training.
- Combined hardware repair expertise with programming skills to improve IoT prototype reliability.
- Gained recognition during the IoT track for strong understanding of embedded hardware and software interaction.